

Role-Based Bus Ticket Reservation and Management System Using Spring Boot and RESTful Architecture

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Abstract—Traditional bus ticket reservation systems often rely on manual counter-based booking processes or loosely integrated digital platforms, resulting in long queuing times, booking conflicts, inefficient seat inventory management, and limited real-time visibility for passengers and operators. This paper presents the design and implementation of a role-based Bus Ticket Reservation and Management System developed using Spring Boot and RESTful architectural principles. The system follows a layered architecture that separates controller, service, repository, and data transfer layers to enhance maintainability and extensibility. Core business rules including seat availability validation, prevention of double booking, role-restricted operations, and controlled booking status transitions are enforced at the service layer with declarative transaction management ensuring data consistency. A key design decision implements atomic seat allocation during booking confirmation, preventing overbooking and resource conflicts. Functional validation confirms the correct enforcement of business rules and transactional behavior.

Index Terms—Bus Ticket Reservation, Spring Boot, RESTful API, Role-Based Access Control, Transaction Management, Seat Allocation, Booking Management System, Layered Architecture, ACID Properties, Database Consistency

I. INTRODUCTION

Public transportation management systems play a vital role in improving passenger experience and operational efficiency in the transit sector. Traditional ticket booking methods such as physical counter reservations or phone-based scheduling often lead to long waiting periods, human errors, booking overlaps, and inefficient utilization of transit resources.

Manual ticketing lacks centralized control and real-time seat availability updates, causing frequent conflicts when multiple passengers attempt to book the same seat. Although digital bus reservation systems have emerged to address these issues, many existing implementations focus primarily on user interfaces while neglecting backend architectural robustness.

This paper presents a Bus Ticket Reservation and Management System developed as an academic prototype to demonstrate modern software engineering practices using Spring Boot and RESTful APIs. The system manages the complete ticket lifecycle from search and booking requests to confirmation and cancellation while enforcing strict business rules and ensuring transactional consistency.

A. Contributions

The key contributions include: (1) layered architectural design using Controller-Service-Repository pattern, (2) comprehensive server-side validation framework, (3) transaction-coordinated booking workflow ensuring ACID properties, (4) RESTful API design with proper endpoint organization, (5) dual-role system architecture with role-based access control, and (6) dynamic seat inventory management with real-time availability tracking.

II. LITERATURE REVIEW

A. Early Web-Based Solutions

Early academic efforts focused on replacing manual ticketing with online forms and database-backed storage. These implementations demonstrated improved processing speed but suffered from procedural designs lacking service-layer abstraction, limiting maintainability and testability.

B. Modern Framework Implementations

Recent studies adopted modern web frameworks, demonstrating responsive frontend design and REST API integration. However, many acknowledged limitations in transaction management, as NoSQL databases' eventual consistency models introduced integrity risks during concurrent booking updates.

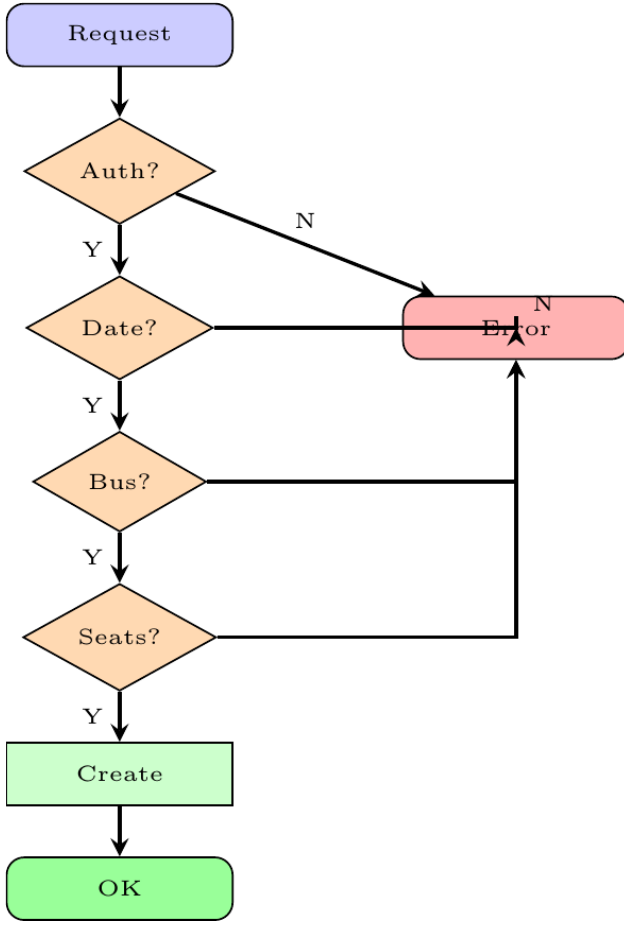


Fig. 1. Booking workflow with validation

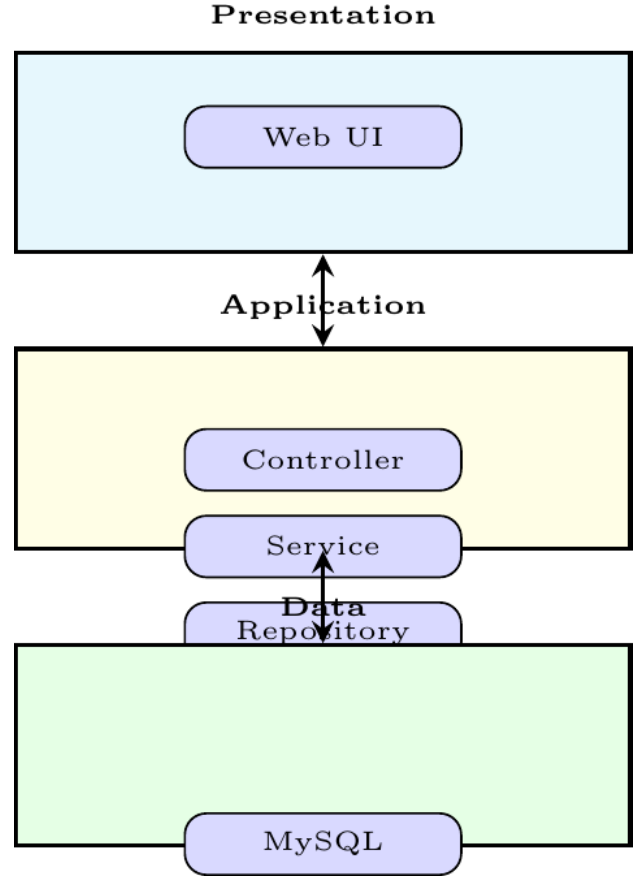


Fig. 2. Three-tier system architecture

C. Identified Gaps

Analysis reveals recurring gaps: (1) lack of clear architectural separation, (2) validation enforced only at UI level, (3) non-atomic booking confirmation workflows, (4) absent overlapping seat allocation detection, (5) limited administrative controls, and (6) inconsistent cancellation handling. Our system addresses these through strict architectural separation, comprehensive server-side validation, and explicit transaction boundaries.

III. SYSTEM ARCHITECTURE

The proposed system adopts a three-tier client-server architecture separating presentation, application logic, and data persistence concerns.

A. RESTful API Design

The application exposes REST endpoints across domain-specific controllers: UserController (registration/authentication), BusController (fleet management/search), BookingController (booking lifecycle), and AdminController (system configuration). Endpoints follow resource-oriented

naming with appropriate HTTP verbs and JSON data interchange format.

B. Data Model and Entity Relationships

The data model comprises four core JPA entities: User (authentication credentials and role designation), Bus (fleet information including capacity and pricing), Booking (reservation details with foreign keys to User and Bus), and Route (route information supporting journey planning). Entity relationships employ JPA annotations with proper cascade configurations and foreign key constraints.

C. Three-Tier Architecture

Presentation Tier: A web-based interface provides bus search, ticket booking, and administrative operations. Communication occurs through RESTful HTTP APIs.

Application Tier: Implemented using Spring Boot 3.x, encapsulating business logic, validation rules, and transaction coordination. Internally follows Model-View-Controller pattern with Controller, Service, and Repository sub-layers.

Data Tier: MySQL relational database managed through Hibernate ORM provides persistent storage with ACID transaction guarantees and referential integrity enforcement.

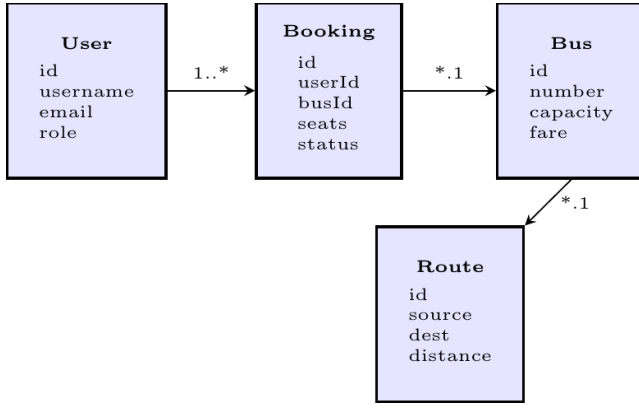


Fig. 3. Core entity relationships

IV. SYSTEM DESIGN AND IMPLEMENTATION

A. Authentication and Role Management

The authentication module implements user registration and login functionality. Registration validates username and email uniqueness before creating User entity with assigned role (USER, ADMIN). The system implements role-based access control at the service layer, ensuring regular users can search buses and manage their bookings, while administrators access extended fleet management and system configuration capabilities.

B. Bus Management Module

Bus registration establishes essential information including identification number, route details, timing information, capacity configuration, and pricing.

Bus Search and Filtering: The system implements advanced search functionality enabling passengers to locate suitable transportation options. Search parameters include source location, destination location, travel date, and seat requirements. The search operation executes database queries identifying buses matching specified criteria with sufficient seat availability.

Administrative Operations: Administrators access extended capabilities including adding new buses, updating information, adjusting seat capacity, modifying fares, controlling operational status, and removing buses from inventory. All modifications execute within transactional boundaries to maintain data consistency.

C. Booking Lifecycle and Validation

Booking Creation Process: Ticket booking submission initiates multi-stage validation: (1) user authentication verification, (2) temporal validation ensuring future travel date, (3) bus existence confirmation, (4) seat availability verification, (5) passenger information validation, (6) double booking prevention, and (7) amount calculation. Upon successful validation, the system creates a Booking entity with CONFIRMED status.

Seat Availability Management: Seat availability tracking implements atomic updates coordinated within database transactions. The booking confirmation process executes operations atomically: retrieve Bus entity, validate requested seats, create Booking record, decrement available seats, and save both entities. The `@Transactional` annotation establishes transaction boundaries ensuring all operations succeed or fail together.

```

@Transactional
public Booking createBooking(
    BookingRequest request) {
    Bus bus = busRepository
        .findById(request.getBusId())
        .orElseThrow(() ->
            new ResourceNotFoundException(
                "Bus not found"));

    if (bus.getAvailableSeats() <
        request.getNumberOfSeats()) {
        throw new InsufficientSeatsException(
            "Only " + bus.getAvailableSeats()
            + " seats available");
    }

    Booking booking = new Booking();
    booking.setBus(bus);
    booking.setNumberOfSeats(
        request.getNumberOfSeats());
    booking.setBookingStatus(
        BookingStatus.CONFIRMED);

    bus.setAvailableSeats(
        bus.getAvailableSeats() -
        request.getNumberOfSeats());

    busRepository.save(bus);
    return bookingRepository.save(booking);
}
  
```

D. Transaction-Coordinated Cancellation Workflow

Booking cancellation demonstrates transaction management principles ensuring atomic seat release and status updates. The cancellation service method executes: retrieve Booking entity, validate CONFIRMED status, retrieve associated Bus entity, update booking status to CANCELLED, restore available seats, and save modified entities. The transactional boundary ensures operations complete atomically.

E. Administrative System Configuration

Route Management: Administrators configure bus routes defining source-destination pairs, distance, journey duration, and operational status.

Dynamic Fare Adjustment: The system supports bulk fare updates, percentage-based adjustments, and absolute fare modifications for individual buses. Fare changes apply immediately to new bookings while preserving historical pricing.

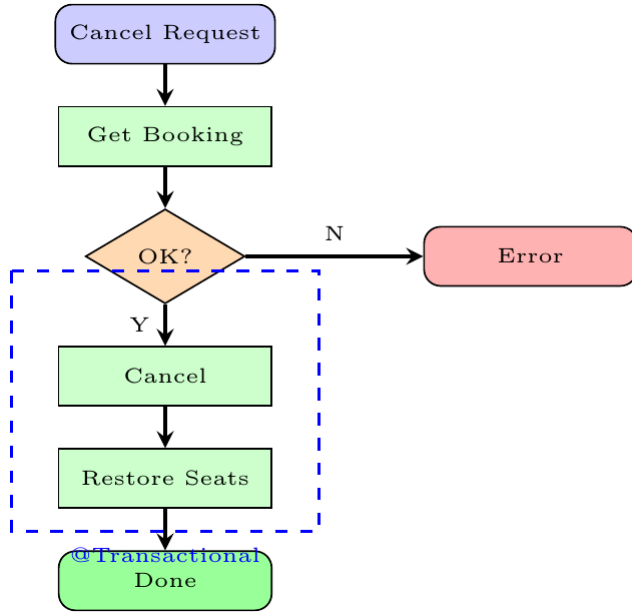


Fig. 4. Cancellation workflow

Seat Capacity Management: Bus capacity modifications enable administrators to increase or decrease total seat count while preventing capacity reductions that would invalidate existing confirmed bookings.

F. Data Transfer Objects and Exception Handling

The implementation employs DTOs separating API contracts from database entities. Request DTOs define client-submitted data with Jakarta Validation annotations. Response DTOs control data exposure, excluding sensitive fields.

Exception handling employs custom exception classes: `InvalidBookingException`, `ResourceNotFoundException`, `InsufficientSeatsException`, `BookingConflictException`, and `UnauthorizedOperationException`. Controllers catch service layer exceptions and construct error response objects with appropriate HTTP status codes, descriptive messages, and timestamps.

G. Comparative Analysis

Compared to conventional manual ticket booking processes, the implemented system provides measurable improvements: reduced booking time (from minutes to seconds), real-time availability visibility, eliminated double booking through automated validation, centralized fleet management reducing operational overhead, comprehensive audit trail enabling accountability, and instant seat release improving inventory utilization.

V. RESULTS AND PERFORMANCE EVALUATION

A. Functional Validation Results

Functional validation was conducted using scenario-based API testing covering typical workflows and error conditions.

Core Validation Rules: Ten core validation rules were verified: (1) date range correctness, (2) seat availability verification, (3) double booking prevention, (4) bus existence

validation, (5) user authentication, (6) status transition enforcement, (7) role-based authorization, (8) booking ownership, (9) passenger information completeness, and (10) amount calculation accuracy. All validation rules demonstrated correct enforcement across standard scenarios and edge cases.

Transaction Management Verification: Transaction management was tested by simulating faults during booking and cancellation workflows. Results confirmed atomicity (complete rollback on failures), consistency (valid database state maintained), isolation (serialized concurrent bookings), and durability (persistence across restarts).

B. System Performance Characteristics

Read Operations: Simple retrieval operations exhibit low latency (100-200ms average) due to indexed database access and minimal business logic processing.

Write Operations: Transactional workflows incur higher overhead (200-300ms average) due to multi-stage validation and coordinated database updates within transactional boundaries.

Scalability Considerations: The system follows Spring Boot's default embedded server configuration, making it suitable for small to medium bus operators under typical usage patterns.

VI. DISCUSSION

A. Design Validation and Key Observations

The implementation successfully demonstrates several software engineering principles. The Controller-Service-Repository pattern provides clear separation of concerns with distinct layer responsibilities, improving code organization and maintainability. Declarative `@Transactional` annotations cleanly handle atomic operations without manual transaction demarcation. The DTO pattern successfully decoupled API contracts from persistence models, enabling independent schema evolution.

B. Implementation Challenges

Several challenges emerged: (1) formulating accurate seat availability queries under concurrent access required careful consideration of transaction isolation levels, (2) implementing comprehensive multi-layer validation without code duplication required thoughtful service layer design, (3) determining appropriate scope for administrative overrides while maintaining system integrity, and (4) ensuring atomic seat release and status updates during cancellations.

C. Limitations and Constraints

Despite functional correctness, the system has limitations: basic authentication without Spring Security integration, stubbed payment processing and refund workflows, absent email/SMS notifications, single-operator deployment only, missing advanced features (seat selection, dynamic pricing, loyalty programs), and unoptimized performance (no caching, pagination, or query optimization).

VII. CONCLUSION AND FUTURE WORK

A. Conclusion

This paper presented a Bus Ticket Reservation and Management System prototype demonstrating modern software engineering practices using Spring Boot and RESTful architecture. The implementation successfully addresses common challenges through layered architectural design, comprehensive server-side validation, transaction-coordinated workflows, role-based access control, RESTful API design, and dynamic administrative capabilities. The atomic seat allocation approach combined with transactional booking workflows prevents overbooking while maintaining system integrity.

B. Future Work

Several enhancements would move this prototype toward production readiness: (1) Spring Security integration with JWT authentication, (2) payment gateway integration supporting multiple payment methods, (3) email/SMS/push notification services, (4) visual seat selection and booking modification capabilities, (5) GPS-based real-time tracking, (6) demand-based dynamic pricing algorithms, (7) comprehensive analytics dashboard, (8) multi-operator platform support, (9) performance optimization with Redis caching and containerization, (10) comprehensive automated testing suite, and (11) native mobile applications. These enhancements would transform the academic prototype into a production-ready platform suitable for real-world deployment.

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